Cancer Biology Fundamentals

Cell Cycle Dysregulation, Oncogenes, and Tumor Suppressor Genes

Doc #01 of 15	**Domain:** Oncology & Cancer Treatment	**Audience:** Medical Professionals

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1. Overview of Normal Cell Division

Normal somatic cells progress through a tightly regulated cell cycle comprising G1 (gap 1), S (DNA synthesis), G2 (gap 2), and M (mitosis) phases. Cyclins and cyclin-dependent kinases (CDKs) act as molecular switches at each checkpoint. CDK4/6–cyclin D complexes phosphorylate retinoblastoma protein (Rb), releasing E2F transcription factors and committing the cell to S-phase entry. The spindle assembly checkpoint (SAC) at the metaphase-to-anaphase transition ensures faithful chromosome segregation by monitoring kinetochore-microtubule attachment.

> ***KEY POINT: Dysregulation of any single checkpoint can be sufficient to initiate neoplastic transformation if compounded by additional mutations.***

2. Hallmarks of Cancer

Hanahan and Weinberg's landmark framework (originally 2000, revised 2011 and 2022) identifies the following acquired capabilities shared by most human cancers:

- Sustaining proliferative signaling (constitutive activation of growth-factor pathways)
- Evading growth suppressors (loss of Rb, p16INK4a, or APC function)
- Resisting cell death (Bcl-2 overexpression, caspase inactivation)
- Enabling replicative immortality (telomerase re-activation via TERT)
- Inducing angiogenesis (VEGF upregulation, HIF-1α stabilisation)

- Activating invasion and metastasis (E-cadherin loss, matrix metalloprotease secretion)
- Reprogramming energy metabolism (Warburg effect – aerobic glycolysis)
- Evading immune destruction (PD-L1 upregulation, MHC-I downregulation)
- Tumour-promoting inflammation (IL-6, TNF-α, COX-2 in the microenvironment)
- Genome instability and mutation (MMR deficiency, BRCA1/2 loss)

3. Oncogenes & Proto-oncogenes

Proto-oncogenes encode proteins that positively regulate cell growth. Point mutations, chromosomal translocations, or gene amplification convert them into dominantly acting oncogenes.

Oncogene	Protein Class	Activation Mechanism	Associated Malignancy
KRAS	GTPase	Point mutation G12D/V/C	PDAC, NSCLC, CRC
MYC	Transcription factor	Amplification / translocation	Burkitt lymphoma, breast
HER2 (ERBB2)	RTK	Amplification	Breast, gastric
BCR-ABL1	Fusion kinase	t(9;22) translocation	CML, ALL
BRAF	Serine/threonine kinase	V600E point mutation	Melanoma, thyroid
PIK3CA	Lipid kinase	Gain-of-function mutation	Breast, endometrial

4. Tumor Suppressor Genes

Tumor suppressors act as 'brakes' on proliferation. Loss-of-function via the Knudson 'two-hit' hypothesis is the predominant mechanism. Germline mutations in one allele predispose to cancer; somatic loss of the second allele triggers malignant transformation.

- TP53 — mutated in ~50% of all cancers; governs G1/S and G2/M checkpoints, apoptosis, and senescence
- RB1 — paradigmatic two-hit gene; loss drives retinoblastoma, SCLC, and osteosarcoma
- BRCA1/2 — homologous recombination repair; germline variants increase lifetime breast/ovarian cancer risk to 70-85%
- APC — regulates Wnt/β-catenin; biallelic loss is the initiating event in ~80% of sporadic CRC
- PTEN — PIP3 phosphatase opposing PI3K; frequently silenced in glioblastoma, endometrial cancer

5. DNA Damage & Repair Pathways

Cells deploy multiple overlapping DNA repair systems. Defects in these pathways simultaneously drive carcinogenesis and create exploitable vulnerabilities for therapy (synthetic lethality).

Repair Pathway	Lesion Type	Key Proteins	Therapeutic Relevance
BER	Oxidative base damage	OGG1, APEX1, PARP1	PARP inhibitors in BRCA-deficient tumours

NER	Bulky adducts, UV dimers	XPC, ERCC1, XPD	ERCC1 predicts platinum sensitivity
MMR	Replication mismatches	MLH1, MSH2, MSH6	MSI-H status predicts ICI response
HR	DSBs (high fidelity)	BRCA1, BRCA2, RAD51	PARP inhibitor sensitivity
NHEJ	DSBs (error-prone)	Ku70/80, DNA-PKcs	Radiosensitisation target

6. Clinical Relevance

Understanding cancer biology directly informs therapeutic strategy. Molecular profiling of tumours guides selection of targeted agents (e.g., EGFR inhibitors for exon 19/21-mutant NSCLC), identifies patients likely to benefit from immunotherapy (high TMB, dMMR), and enables early detection through liquid biopsy circulating tumour DNA (ctDNA) assays.

> **■ CLINICAL NOTE: All molecular findings must be interpreted in the context of validated assay platforms and correlated with histopathological diagnosis before treatment decisions are made.**